Probability-3.0.27

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Question

The time between successive failures (in hours) of a side discharge loader operating in a mechanised underground coal mine are as follows:

$$62, 58, 54, 50, 52, 60, 58, 57, 50, 53$$

If the failure data follow an exponential distribution, then reliability of the equipment for a period of 50 hours is:

1) 0.25 2) 0.40 3) 0.60 4) 1.00

Parameter Estimation

For an exponential distribution, we first find the Mean Time Between Failures (μ):

$$\mu = \frac{\sum t_i}{n} = \frac{554}{10} = 55.4 \text{ hours}$$

As the given is a exponential distribution, failure rate (λ) is the reciprocal of the μ:

$$\lambda = \frac{1}{\mu} = \frac{1}{55.4} \approx 0.01805 \text{ failures/hour}$$

PDF and CDF Definitions

Probability Density Function (PDF): Represents the failure density at time t.

$$f(t) = \lambda e^{-\lambda t}$$

Cumulative Distribution Function (CDF): Represents the probability that failure occurs *before* time t.

$$F(t) = P(T \leq t) = \int_0^t \lambda e^{-\lambda x} dx = 1 - e^{-\lambda t}$$

Reliability Calculation

Reliability $R(t)$ is the probability of survival beyond time t. It is the complement of the CDF:

$$R(t) = 1 - F(t) = e^{-\lambda t}$$

Substitute $t = 50$ and $\lambda = 0.01805$:

- $R(50) = e^{-(0.01805)(50)}$
- $R(50) = e^{-0.9025}$
- $R(50) \approx 0.4055$

Final Conclusion

Result

The reliability of the side discharge loader for a 50-hour period is approximately **0.40**.

Correct Option: (b)

- $P(\text{Failure before 50h}) \approx 60\%$ (CDF)
- $P(\text{Survival past 50h}) \approx 40\%$ (Reliability)

Python Code for Plot

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters from the problem
data = [62, 58, 54, 50, 52, 60, 58, 57, 50, 53]
mtbf = np.mean(data)
lambd = 1 / mtbf
t_val = 50

# Time range for plotting
t = np.linspace(0, 200, 1000)

# CDF: F(t) = 1 - exp(-lambda * t)
cdf = 1 - np.exp(-lambd * t)
```

Python Code for Plot

```
# Reliability: R(t) = exp(-lambda * t)
reliability = np.exp(-lambd * t)

# Calculations at t = 50
cdf_50 = 1 - np.exp(-lambd * t_val)
rel_50 = np.exp(-lambd * t_val)

# Create the plot
plt.figure(figsize=(10, 6))
plt.plot(t, cdf, color='green', lw=2, label=r'CDF
    $F(t) = 1 - e^{-\lambda t}$ (Failure Prob.)')
plt.plot(t, reliability, color='orange', lw=2,
   label=r'Reliability $R(t) = e^{-\lambda t}$ (
   Survival Prob.)')
plt.axvline(t_val, color='red', linestyle='--',
   alpha=0.7)
```

Python Code for Plot

```
plt.scatter([t_val, t_val], [cdf_50, rel_50],
    color='red', zorder=5)

# Annotate values at t=50
plt.annotate(f'CDF(50) $\\approx$ {cdf_50:.2f}',
    xy=(t_val, cdf_50), xytext=(t_val+10, cdf_50
    +0.05),
                arrowprops=dict(facecolor='black',
                    shrink=0.05, width=1, headwidth
                    =5))
plt.annotate(f'R(50) $\\approx$ {rel_50:.2f}', xy
    =(t_val, rel_50), xytext=(t_val+10, rel_50
    -0.1),
                arrowprops=dict(facecolor='black',
                    shrink=0.05, width=1, headwidth
                    =5))
```

Python Code for Plot

```
plt.title('CDF vs. Reliability Function',
    fontsize=14)
plt.xlabel('Time (hours)', fontsize=12)
plt.ylabel('Probability', fontsize=12)
plt.legend()
plt.grid(True, linestyle=':', alpha=0.6)

plt . tight_layout ()
plt . show ()
```

Plot

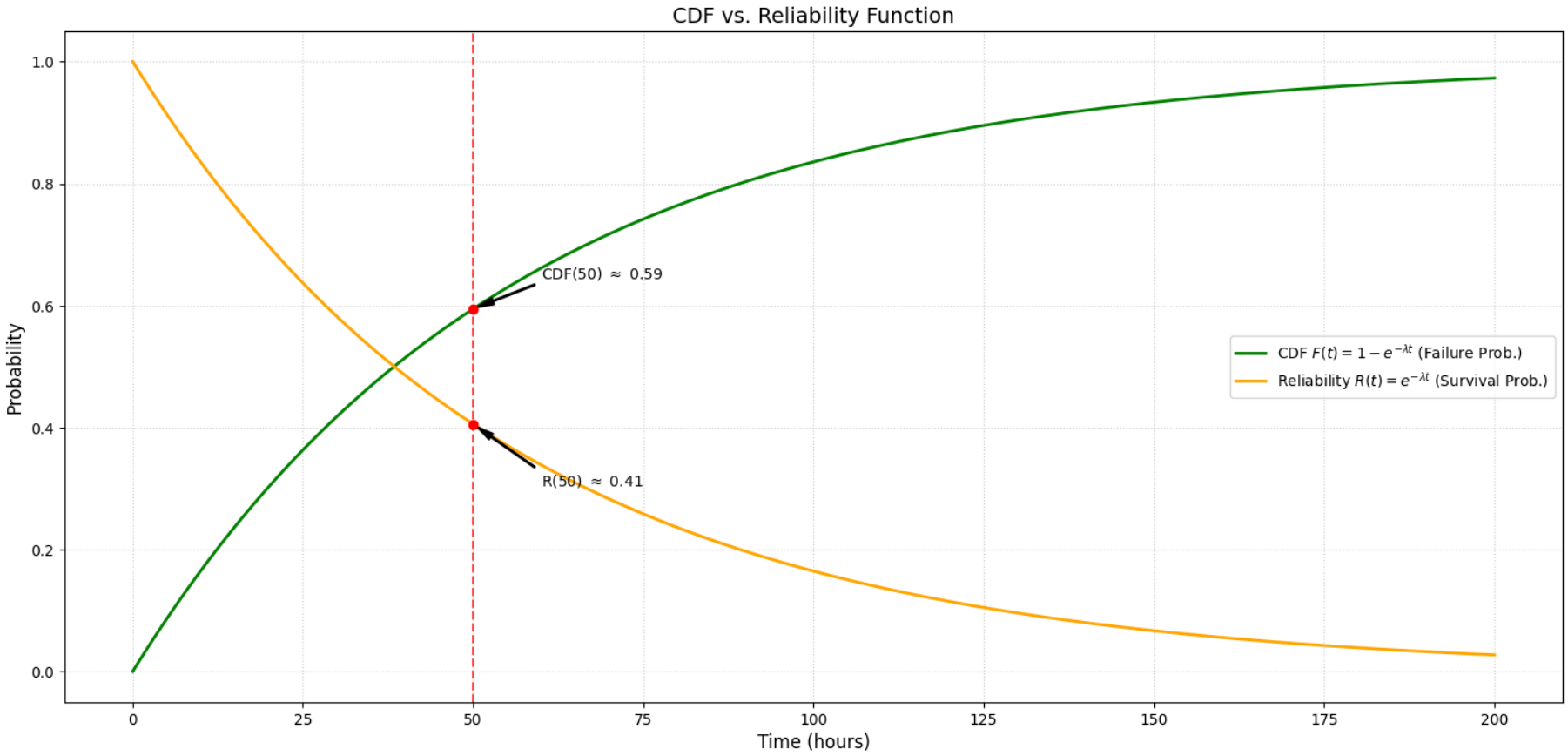

Figure: Python plot